

Chapter 1

Environmental Mitigation Costs Calculation Documentation

1.1 Introduction

The `environmental_mitigation.py` script calculates environmental mitigation costs for a transmission line project. It computes base construction/restoration costs and wetland/habitat credit purchases across different construction types and terrain types, applying uplift factors for effective acreage.

1.2 Method Overview

Canonical notation. The canonical variable names and equations (including $A_{\text{terrain,habitat}}^{\text{effective impact}}$ for effective acres of habitat impact per terrain, and $C_{\text{credits,habitat}}$ as a sum over habitat terrains using that per-terrain acreage) are defined in `COST_METHODODOLOGY.md` under “Environmental mitigation (1.c)”. This document summarizes the implementation in code.

1.2.1 Purpose

This script calculates environmental mitigation costs by:

1. Calculating ROW acres by terrain type
2. Applying uplift factor to determine effective acres (TCE - Total Cost Equivalent)
3. Computing base mitigation/restoration costs per effective acre
4. Calculating wetland credit purchases for impacted wetland acres
5. Calculating habitat credit purchases for impacted natural habitat acres
6. Applying AFUDC capitalization for regulatory perspective
7. Computing present values for societal perspective

1.2.2 Calculation Approach

The environmental mitigation cost calculation follows this structure:

$$\text{Terrain Acres} = \frac{\text{Terrain Miles} \times 5280 \times \text{ROW Width (feet)}}{43560} \quad (1.1)$$

$$\text{Effective Acres} = \text{Base Acres} \times \text{Uplift Factor} \quad (1.2)$$

$$\text{Base Cost} = \sum_{\text{terrains}} \text{Cost/Acre} \times \text{Effective Acres} \quad (1.3)$$

$$\text{Wetland Credits} = \text{Wetland Cost/Acre} \times \text{Ratio} \times \text{Wetland Impacted Acres} \quad (1.4)$$

$$\text{Habitat Credits} = \text{Habitat Cost/Acre} \times \text{Ratio} \times \text{Habitat Impacted Acres} \quad (1.5)$$

$$\text{Total} = \text{Base Cost} + \text{Wetland Credits} + \text{Habitat Credits} \quad (1.6)$$

The uplift factor accounts for additional mitigation requirements beyond the physical ROW footprint (e.g., buffer zones, restoration areas).

1.2.3 Inputs and Outputs

Inputs:

- Project technical details (construction type, AC/DC, capacity, etc.)
- Physical details (terrain distribution)
- ROW width for project category
- Environmental mitigation parameters (base costs, credit costs, credit ratios, uplift factor)
- Financing parameters (WACC, inflation, AFUDC rate)
- Timeline (delay years, construction years)

Outputs:

- Nominal costs (base mitigation, wetland credits, habitat credits)
- AFUDC capitalized costs (regulatory perspective)
- Present value costs (societal perspective)
- Acreage metrics (base acres, effective acres, impacted acres)

1.2.4 Integration with Other Scripts

This script integrates with:

- `smart_loaders.py` — `load_project_technical_details`, `load_physical_details`, `load_row_width`, `load_environmental_mitigation`, `load_financing_details`, `load_cost_timing_patterns`, `load_afudc_config`, `get_physical_data_raw`, `get_financing_data_raw`, `load_terrain_miles`
- `calculation_utils.py` — `build_category_string`, `normalize_construction_type_for_yaml`, `miles_to_acres`
- `financial_utils.py` — `calculate_present_value`, `calculate_afudc_capitalized_cost`, `validate_discount_rate`, `calculate_construction_start_year`, `load_afudc_setup`
- `path_config.py` — `YAMLS_DIR`
- `smart_output.py` — `CTCCOutputManager` (`add_environmental_mitigation`, `write_batch_summary`)

1.3 Key Functions

1.3.1 `calculate_environmental_mitigation_costs(em_yaml, category, terrain_miles_dict, row_width_feet, reconductoring)`

Calculates environmental mitigation costs including base mitigation and credit purchases.

Parameters:

- `em_yaml` (dict): Loaded environmental mitigation YAML data
- `category` (str): Project category identifier
- `terrain_miles_dict` (dict): Dictionary mapping terrain type to miles
- `row_width_feet` (float): Right-of-way width in feet
- `reconductoring` (bool): If True, sets wetland and habitat credits to zero. Reconductoring projects use existing ROW and structures, so they don't create new permanent environmental impacts requiring mitigation credits. Base mitigation costs still apply for temporary construction impacts.

Returns:

- Dictionary containing:
 - `base_cost` (float): Base mitigation/restoration cost
 - `wetlands_credits` (float): Wetland credit purchase cost
 - `habitat_credits` (float): Habitat credit purchase cost
 - `total_credits` (float): Sum of wetland and habitat credits
 - `total` (float): Total mitigation cost

- uplift_factor_applied (float): Uplift factor used
- total_base_acres (float): Base ROW acres
- total_effective_acres (float): Effective acres with uplift
- wetland_impacted_acres (float): Wetland acres impacted
- habitat_impacted_acres (float): Habitat acres impacted

Key Logic:

```

1 construction_type = category.split("/") [0]
2 yaml_construction_type = normalize_construction_type_for_yaml(
   construction_type, context="environmental")
3 for terrain, miles in terrain_miles_dict.items():
4     if miles > 0:
5         terrain_acres = miles_to_acres(miles, row_width_feet)
6         acres_by_terrain[terrain] = terrain_acres
7         total_base_acres += terrain_acres
8         effective_acres = terrain_acres * uplift_factor
9         cost_per_acre = base_costs.get(yaml_construction_type, {}).
           get(terrain, 0.0)
10        base_cost += cost_per_acre * effective_acres
11 wetland_impacted_acres = acres_by_terrain.get("wetland", 0.0) *
   uplift_factor
12 habitat_impacted_acres = sum(acres_by_terrain.get(t, 0.0) for t in
   habitat_terrains) * uplift_factor
13 wetlands_credits = wetland_cost_per_acre * wetland_ratio *
   wetland_impacted_acres
14 habitat_dict = credits.get("habitat", {})
15 ratios_habitat = ratios.get("habitat", {})
16 default_cost, default_ratio = habitat_dict.get("default", 0.0),
   ratios_habitat.get("default", 1.0)
17 habitat_credits = 0.0
18 for terrain in habitat_terrains:
19     terrain_acres = acres_by_terrain.get(terrain, 0.0)
20     if terrain_acres > 0:
21         effective_acres = terrain_acres * uplift_factor
22         cost_per_acre = habitat_dict.get(terrain, default_cost)
23         ratio_val = ratios_habitat.get(terrain, default_ratio)
24         habitat_credits += cost_per_acre * ratio_val *
           effective_acres
25 if reconductoring:
26     wetlands_credits = 0.0
27     habitat_credits = 0.0

```

1.3.2 main()

Main execution function that orchestrates the environmental mitigation cost calculation and output.

Workflow:

1. Loads project technical details; constructs category via `build_category_string`; loads ROW width via `load_row_widths(category)`; loads terrain via `load_terrain_miles()`; loads `em_yaml` via `load_environmental_mitigation()`
2. Calls `calculate_environmental_mitigation_costs(em_yaml, category, terrain_miles, row_width_feet, project_details.reconductoring)`
3. Loads `load_afudc_setup()`; if `apply_afudc`, calculates AFUDC for base and credits separately using `timing_patterns["environmental_mitigation_base"]` and `["environmental_mitigation_credits"]`
4. Credits PV: one-time at `credit_start_year = calculate_construction_start_year(delay_years)`
base PV: `calculate_present_value(annual_base_cost, wacc_real, construction_years, credit_start_year)` (or one-time at `credit_start_year` if `construction_years` is 0)
5. `add_environmental_mitigation(results); write_batch_summary()`

1.4 Example

The following example demonstrates a typical usage of the environmental mitigation cost calculation:

```
1 # Example: 500 MW Overhead AC line
2 # Category: "Overhead/AC/500MW/Advanced Aluminum/NA"
3 #
4 # Inputs (from YAML):
5 #   - ROW width: 200 feet
6 #   - Terrain distribution:
7 #     * Forested: 40 miles
8 #     * Wetland: 10 miles
9 #     * Farmland: 50 miles
10 #   - Uplift factor: 1.5x
11 #   - Base cost: $2000/acre (overhead, forested), $3000/acre (
12     overhead, wetland)
13 #   - Wetland credit: $5000/acre, ratio 2.0:1
14 #   - Habitat credit: $3000/acre, ratio 1.5:1 (in full
15     implementation, cost and ratio can vary by habitat terrain)
16 #   - Construction: 2 years
17 #
18 # Calculation:
19 #   Forested acres = (40 * 5280 * 200) / 43560 = 969.7 acres
20 #   Wetland acres = (10 * 5280 * 200) / 43560 = 242.4 acres
21 #   Farmland acres = (50 * 5280 * 200) / 43560 = 1212.1 acres
22 #
23 #   Effective acres (with 1.5x uplift):
24 #     * Forested: 969.7 * 1.5 = 1454.6 acres
25 #     * Wetland: 242.4 * 1.5 = 363.6 acres
```

```

24 #
25 #   Base cost = (1454.6 * $2000) + (363.6 * $3000) = $4.4M
26 #   Wetland credits = $5000 * 2.0 * 363.6 = $3.6M
27 #   Habitat credits = $3000 * 1.5 * 1454.6 = $6.5M
28 #   Total = $14.5M
29 #
30 # Output:
31 #   - Nominal cost: $14.5M
32 #   - AFUDC capitalized: $15.2M (if AFUDC enabled)
33 #   - Present value: $13.8M (base spread over 2 years, credits
    upfront)

```

Integration Example:

```

1 # When called from ctcc.py:
2 csv_manager = CTCCOutputManager()
3 csv_results = {
4     "total_nominal": results["total"],
5     "total_afudc": total_capitalized if apply_afudc else 0,
6     "total_pv": total_pv,
7     "base_cost_nominal": results["base_cost"],
8     "credits_nominal": results["total_credits"],
9 }
10 csv_manager.add_environmental_mitigation(csv_results)
11 csv_manager.write_batch_summary()

```

1.5 Key Implementation Details

1.5.1 Uplift Factor

The uplift factor (TCE - Total Cost Equivalent) accounts for mitigation requirements beyond the physical ROW footprint. For example, a 1.5x uplift means that for every acre of ROW, 1.5 acres of mitigation is required (including buffer zones, restoration areas, etc.).

1.5.2 Credit Ratios

Credit ratios represent the mitigation requirement. For example, a 2.0:1 ratio for wetlands means that for every acre of wetland impacted, 2 acres of wetland credits must be purchased.

1.5.3 Habitat Terrain Classification

The script identifies habitat-impacted acres by summing natural terrain types: forested, scrubbed.flat, desert.barren, rolling.hills, and mountain. This is a conservative approach that captures all potential habitat impacts. Habitat credit cost and ratio can be specified per terrain in the configuration (with a `default` fallback); the total habitat credit cost is the sum over habitat terrains of $(\text{cost_per_acre} \times \text{ratio} \times \text{effective_acres})$ for each terrain.

1.5.4 Present Value Timing

- **Base mitigation costs:** Spread evenly over construction period (annual payments)
- **Credit costs:** One-time payment at start of construction (end of delay period)

1.5.5 AFUDC Treatment

Base mitigation and credit costs are treated separately for AFUDC calculation, as they may have different timing patterns during construction.

1.5.6 Reconductoring Projects

For reconductoring projects, environmental mitigation credits (wetland and habitat credits) are set to zero. This reflects that reconductoring projects use existing ROW and structures, so they do not create new permanent environmental impacts requiring mitigation credits. However, base environmental mitigation costs still apply to account for temporary construction impacts such as:

- Temporary access roads and staging areas
- Erosion control and sediment management during construction
- Temporary disturbance of existing ROW during conductor replacement

This treatment makes reconductoring economics more realistic by avoiding the large credit costs that would be inappropriate for projects that don't disturb new land.